Mapping the Landscape: Empirical Insights into Research Practices in Computational Linguistics

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Abstract

Computational linguistics, at the intersection of linguistics, informatics, and artificial intelligence, continues to evolve both methodologically and conceptually. The precise boundaries of the field remain indistinct, leaving the question of what constitutes core research open to continual debate. This study presents a systematic empirical examination of computational linguistics scholarship by curating a corpus of 2,000 paper abstracts, annotated with a novel taxonomy to categorize scientific contributions and research types. We introduce an automatic classification task, benchmarking robust baseline models, and apply these models to over 28,000 articles to uncover field-wide patterns. The investigation highlights a marked increase in machine learning-centric work since the 1990s and a parallel decrease in papers focused on linguistic theory or human annotation. Notably, recent years show renewed interest in interdisciplinary contributions, as well as ethical and social aspects of language technologies. By charting these longitudinal trends, our work provides a resource for understanding community standards, shaping future directions, and promoting thoughtful discourse on the evolving identity of computational linguistics.

1 Introduction

> "He who does not know to which port he is sailing, no wind is favorable."
>
> - Seneca

Categorizing research by scientific discipline has several benefits, including bringing together scientists to make progress in a cohesive area of interest. While there is often some broad description of what a scientific discipline constitutes, the nature of a discipline is dynamic and multifaceted, and it can change with time. NLP is a particularly interesting discipline in this regard, not only because of its interdisciplinary nature, drawing on ideas and techniques from computer science, linguistics, social science, etc., but also because fundamental questions such as '*what is NLP research?*' can be contentious. Is it the study and development of algorithms that give machines the ability to respond to and generate language? Is it the study of natural language using computational approaches? Does it cover all research at the intersection of computation and language? Or is it something more narrow?

A compelling way to answer '*what is NLP research?*' is to examine the papers published in NLP conferences and journals. After all, the body of current published research is the best indicator of what a field is and the nature of the field, regardless of how that field may have once been defined or understood.

A key window into the nature of a particular research project is how the authors articulate their contributions. *Contributions* are new scientific achievements attributed to the authors. Roughly speaking, scientific contributions are of two types:
i) those that add to human knowledge, e.g., discovering the structure of DNA; and ii) those that create new and useful artifacts, e.g., a general purpose chat system such as ChatGPT. When authors present their work in scientific papers, they describe their contributions to the research community. We define *contribution statements* as descriptions of these contributions.

In this paper, we propose that automatically extracting, categorizing, and quantitatively

analyzing the contribution statements in the research papers of a field provide key insights into the nature of the field. Additionally, such an effort enables historical (longitudinal) analyses of the field (Kuhn, 1970) and can help researchers identify emerging trends and stay current amid the
rapid proliferation of scientific publications.

We explore this idea concretely and empirically by examining 28,937 NLP papers published between 1974 and 2024. Specifically, we:

1. Introduce a *taxonomy* of contribution types common in NLP papers (§ 3.1).
2. Create a *dataset* NLPContributions comprising of 1,995 NLP research papers with manually annotated contribution statements and contribution types from their abstracts (§ 3.2).
3. Propose a *novel task* to automatically extract and classify contribution statements into contribution types from NLP papers (§ 4.1).
4. Finally, ask (and answer) some preliminary questions on the *nature of NLP research* and how it has changed over the years (§ 5).

We will make all of the data and code freely available through the project webpage.

2 Related Work

Claim and Opinion Summarization. In this area, researchers have explored automated methods to capture diverse perspectives and analyze different angles of claims (Chen et al., 2019). This includes the growing interest in key-point analysis (Bar-Haim et al., 2021). Prior works have proposed neural-network-based and graph-based approaches for claim summarization from newspaper reports and online discussions (Zhao et al., 2022; Inácio and Pardo, 2021). Some research has focused on extracting claims from scholarly documents, particularly in the biomedical field (Achakulvisut et al., 2019; Sosa et al., 2023). While claims in research papers provide declarations to support hypotheses or research questions, contributions present new elements (knowledge and artifacts) that a paper introduces to its field. In this work, we explore methods to extract and analyze contribution statements from NLP research papers.

NLP Scientometrics. In parallel, NLP scientometrics has witnessed significant advancements in recent years as researchers strive to understand the landscape of NLP research and its evolution (Mingers and Leydesdorff, 2015; Chen and Song, 2019). One prominent research direction in NLP scientometrics is the analysis of research trends and patterns (Mohammad, 2020a).

These studies employ bibliometric techniques (Wahle et al., 2022), co-authorship analysis (Mohammad, 2020b), and topic modeling (Jurgens et al., 2018a) to gain insights into the dynamics of the field, identifying influential papers, emerging topics, and research trajectories. Text mining and deep learning techniques have also been utilized in NLP scientometrics to extract information from research papers, create structured datasets, and enable detailed analyses of the interactions among topics (Tan et al., 2017; Salloum et al., 2017; Pramanick et al., 2023). Although some efforts have been made towards discourse relation extraction (Yang and Li, 2018) from scientific papers, as well as studies on the evolution of topics within scientific fields (Hall et al., 2008; Prabhakaran et al., 2016), relatively little is known about the contributions. While NLP scientometrics has concentrated on metadata, largely ignoring the rich contents of the paper, our study delves deeper into analyzing the research paper content. Semantic Content Structuring. Researchers have investigated structured semantic content modeling to improve scientific document searches. D'Souza and Auer (2020) introduced an annotation scheme to identify information units in scientific documents, focusing on artifacts like models, datasets, or baselines linked to a pre-defined set of NLP tasks. Building on this D'Souza et al. (2021) employed this annotation scheme to construct a knowledge graph that connects these artifact information units across NLP tasks. Additionally, researchers have applied deep learning methods to automate the extraction of these information units (Gupta et al., 2021, 2024). It is important to note that these units are not necessarily novel

contributions from the papers they are extracted from. Unlike these efforts, our work broadens the scope by identifying and categorizing contribution statements from research papers without limiting them to specific NLP tasks. Additionally, our approach encompasses contributions that expand knowledge as well as introduce new artifacts. Citation Intent Analysis. A large body of research has focused on understanding the purpose behind citations and developing classification systems for them (Stevens and Giuliano, 1965; Oppenheim and Renn, 1978; Garzone and Mercer, 2000; Teufel et al., 2006; Dong and Schäfer, 2011; Jurgens et al., 2018b). While citation intents signal the purpose of a citation, such as providing background information or making comparisons, contributions differ as they present novel additions that a research paper introduces to its field. However, citation intents may indirectly reflect a paper's

Typ.	Sub-typ.	κ
Knowledge	k-dataset	0.70
	k-language	0.69
	k-method	0.71
	k-people	0.67
	k-task	0.70
Artifact	a-dataset	0.76
	a-method	0.71
	a-task	0.73
Overall Agreement		0.71

Table 1: Inter-annotator Agreement

contributions from the perspective of citing papers; our focus is on contributions as articulated by the authors themselves within their own work.

3 NLPContributions: A Corpus of Contribution Statements

We develop a taxonomy of various types of contributions found in NLP research papers. Using this taxonomy, we annotate contribution statements from the abstracts of NLP research papers. We chose the abstracts as our corpus for annotation because abstracts are uniquely positioned at the beginning of papers and typically contain contribution statements. Moreover, abstracts efficiently summarize the paper, providing the context for understanding contributions, making them particularly suitable text segments to focus on for contribution annotation. Annotating entire papers would substantially escalate annotation efforts, rendering abstracts a more efficient and effective option (Teufel et al., 1999).

3.1 Taxonomy of Contributions

In NLP research, contributions can broadly be divided into *two main types*. We call the first type *artifacts*, which encompasses the development of new or novel resources. NLP research heavily utilizes tools from machine learning, which relies on resources such as new methods or models, datasets - and the novel tasks they enable - all of which are recognized as significant contributions. Consequently, we categorize artifact contributions into three sub-types: *new methods* (a-methods), *new datasets* (a-datasets), and *new tasks* (a-tasks), each distinguished by the specific resource it brings to the field.

We term the second category as *knowledge* contributions that enrich the field with new insights or knowledge. Depending on what these contributions add knowledge to, we further categorize them into five sub-types: *knowledge*

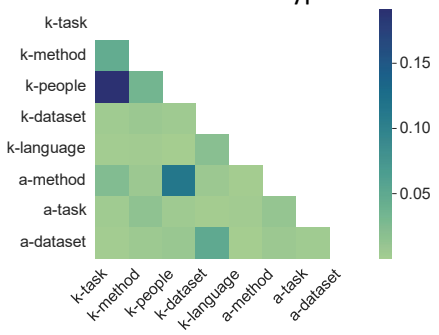


Figure 1: Pointwise Mutual Information (PMI) between labels illustrates the co-occurrence of multiple labels in contribution statements.

about method (k-method), *knowledge about dataset* (k-dataset), *knowledge about task* (ktask), *knowledge about language* (k-language), and *knowledge about people* (k-people). This subcategorization also mirrors the important elements in NLP research.

In Table 3, we provide a detailed description of each type and subtype, with examples of contribution statements from NLP research papers. While we acknowledge that alternative taxonomies may also be possible, we note that our proposed taxonomy is in line with the ACL'23 call for papers[1], which seeks submissions either that conduct analysis (thereby adding *knowledge*) or introduce new resources (thereby adding *artifacts*).

3.2 Curation

Data Preparation. We compile a corpus of abstracts from 1,995 papers published under ACL Anthology using the S2ORC (Lo et al., 2019), a large collection of papers released to support research. We randomly selected these papers, guaranteeing a selection of at least five papers from each year between 1974 and February 2024. The selected papers were published in journals and conferences affiliated with "ACL Events", while those from workshops were excluded. Additionally, we retrieve the metadata (i.e., unique id, title, authors, publication venue, and date) for each selected paper from `anthology.bib`[2].

Annotation. The main annotator is one of the authors of this paper, who has six years of experience in NLP research. Additionally, a PhD student with four years of research experience took part in annotation. We develop an annotation scheme to identify and classify the contribution statements in NLP research papers. We use ontology-oriented annotation guidelines (Refer Appendix A for details) following Liakata et al. (2010). Regular meetings were conducted between the annotators to refine the guidelines as necessary (Klie et al., 2024).

Typ.	Sub-typ.	Prop. (%)
Knowledge	k-dataset	5.1
	k-language	4.0
	k-method	12.6
	k-people	9.2
	k-task	36.1
Artifact	a-dataset	2.2
	a-method	27.2
	a-task	3.6

Table 2: The proportion of Labels in the Annotated Contribution Statements.

Both the annotators annotated 100 papers, ensuring representation from each decade between 1980 and 2024. We assess annotator agreements on these 100 papers. Subsequently, the senior annotator proceeded to annotate the abstracts of the additional 1,895 papers, adhering to the guidelines. Agreement. We measure the inter-annotator agreement (IAA) by comparing the contribution statements from the 100 aforementioned papers annotated by the two annotators under the same contribution labels. All annotations were conducted in Label Studio (Tkachenko et al., 2020-2022). Table 1 shows an average Fleiss' κ of 0.71, comparable to similar works on scholarly documents (Yang and Li, 2018; Lauscher et al., 2022). Further, we observe the error bounds of κ between 0.60 (lower bound) and 0.82 (upper bound) with 95% confidence level ($p < 0.05$).

3.3 Statistics

We highlight three aspects of our dataset: first, the dataset includes abstracts and metadata of $1{,}995$ papers from "ACL Events" in the ACL Anthology (with an average of 2.95 sentences per abstract annotated as contribution statements; on average, an abstract consists of 5.42 sentences), totaling $5{,}890$ annotated contribution statements. Second, we illustrate the distribution of labels across these statements in Table 2. Lastly, we note that 57.6% of the contribution statements received multiple labels. In Figure 1, we illustrate the cooccurrence of different labels through Pointwise Mutual Information (PMI) scores. We observe a high co-occurrence between the labels *k-people* and *k-task*, likely because authors often explain how NLP tasks yield insights into humans or society. We divide our dataset into train-val-test (70-15-15) split at the paper level to maintain

[1] https://tinyurl.com/mpdkmzkj

[2] https://aclanthology.org/anthology.bib.gz

consistency in our experiments and prevent information leakage.

Type	Sub-type	Description	Example
Knowledge	k-dataset	Describes new knowledge about datasets, such as their new properties or characteristics.	"Furthermore, our thorough analysis demonstrates the average distance between aspect and opinion words are shortened by at least 19% on the standard SemEval Restaurant14 dataset." – Zhou et al. (2021)
	k-language	Presents new knowledge about language, such as a new property or characteristic of language.	"In modern Chinese articles or conversations, it is very popular to involve a few English words, especially in emails and Internet literature." – Zhao et al. (2012)
	k-method	Describes new knowledge or analysis about NLP models or methods (which predominantly draw from Machine Learning).	"Different generative processes identify specific failure modes of the underlying model." – Deng et al. (2022)
	k-people	Presents new knowledge about people, humankind, society, or human civilization.	"Combating the outcomes of this infodemic is not only a question of identifying false claims, but also reasoning about the decisions individuals make." – Pacheco et al. (2022)
	k-task	Describes new knowledge about NLP tasks.	"We show that these bilingual features outperform the monolingual features used in prior work for the task of classifying translation direction." – Eetemadi and Toutanova (2014)
Artifact	a-dataset	Introduces a new NLP dataset (i.e., textual resources such as corpora or lexicon).	"We present a new corpus of Weibo messages annotated for both name and nominal mentions." – Peng and Dredze (2015)
	a-method	Introduces or proposes a new or novel NLP method or model (primarily to solve NLP task(s)).	"The paper also describes a novel method, EXEMPLAR, which adapts ideas from SRL to less costly NLP machinery, resulting in substantial gains both in efficiency and effectiveness, over binary and n-ary relation extraction tasks." – Mesquita et al. (2013)
	a-task	Introduces or proposes a new or novel NLP task (i.e., welldefined NLP problem).	"We formulate a task that represents a hybrid of slot-filling information extraction and named entity recognition and annotate data from four different forums." – Durrett et al. (2017)

Table 3: Description of the taxonomy on NLP research contributions with examples.

4 Identifying Contribution Statements and Contribution Types

We introduce the novel task of automatically detecting and categorizing contribution statements from NLP research papers. We use `NLPContributions` and benchmark multiple models to evaluate their performance on this task.

4.1 Task Definition

The task involves two steps: detecting contribution statements and subsequently categorizing them by types. We model it end-to-end as a multi-label extension of multi-class classification where, given a statement, the objective is to assign it types and subtypes if it qualifies as a contribution; otherwise, assign Null. Formally, given a statement S, and a set of n labels $L = [l_1, l_2, ..., l_n]$, the task is to predict a subset of these labels $Y = [y_1, y_2, ..., y_n]$ associated with S, where $y_j = 1$, if l_j is associated with S, and 0 otherwise.

4.2 Methods

In our study, we explore two methods. The first method involves utilizing pre-trained language models (PLMs) that are further fine-tuned using the training split of `NLPContributions`. Second, we use Large Language Models (LLMs) and utilize prompting techniques for our task (Refer to Appendix C for the prompting details). We use the *binary relevance* (Read et al., 2011) for the task, treating each label as an independent binary classification problem. This avoids overfitting by not depending on previous label combinations and allows for flexible modifications to the label set without affecting other parts of the model.

PLMs. We start our study with BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), which are general-purpose pre-trained language models. Moving further, we use BiomedBERT (Gu et al., 2021) and SciBERT (Beltagy et al., 2019), which are pre-trained on scientific texts. Additionally, we experiment with Flan-T5 (Chung et al., 2022), which is pre-trained over a collection of 1,836 finetuning tasks. We also implement a random baseline that assigns labels to sentences with a uniform random probability.

LLMs. Addressing the task through prompting, we use GPT-3.5-Turbo and GPT-4-Turbo (OpenAI, 2023), which are instruction-following large language models fine-tuned with Reinforcement Learning from Human Feedback (RLHF).

Additionally, we use the open-source LLaMA-38B (Meta, 2023) model, which has been trained on over 15 trillion tokens gathered from publicly available domains.

4.3 Training and Evaluation

During fine-tuning the pre-trained language models, we use a grid search across various epochs $e \in \{1,2,3,4,5\}$ and learning rates $lr \in \{1 \cdot 10^{-4}, 5 \cdot 10^{-4}, 1 \cdot 10^{-5}\}$, using a batch size of 32. For prompting, we start with a zero-shot setting and gradually progress to a five-shot, respecting the context length limitations of the models. We repeat each experiment three times and observe the variance < 0.02 for all of the models.

Following Uma et al. (2021), for multi-label classification, we use label-based evaluation (macro-averaged precision, recall, and F1-score), which assesses performance on a per-label basis and then aggregates scores across all labels. We avoid label-set-based evaluation, also known as the exact match measure, because it does not effectively account for the sparsity characteristic of multi-labeling, often missing nuanced label variations.

4.4 Results and Discussion

Table 4 shows the results. We observe that SciBERT outperformed other fine-tuned pretrained language models, likely due to its pretraining on the S2ORC dataset, a collection of scholarly documents. Additionally, we note that

Setting	Model	P	R	F1
	Random	0.19	0.17	0.17
Finetuning	BERT	0.31	0.50	0.38
	PubMedBERT	0.64	0.59	0.60
	SciBERT	0.81	0.80	0.80
	Flan-T5	0.79	0.78	0.78
	RoBERTa	0.33	0.50	0.40
Prompting	GPT-3.5-Turbo	0.75	0.71	0.73
	GPT-4-Turbo	0.80	0.80	0.80
	LLaMA-3	0.60	0.56	0.53

Table 4: Performance of different models for `contribution` statement classification.

GPT-4-Turbo's performance is on par with finetuned SciBERT. Hence, for sustainability and costefficiency reasons, we have chosen to use SciBERT for subsequent analyses. We tested five different prompt variants and recorded the most effective ones in Table 7 (Appendix C). We also found that LLM performance decreased when prompts included titles or entire abstracts, likely because titles may not accurately represent contributions and LLMs are optimized for data with fixed context lengths. All reported results are statistically significant (McNemar's $p < 0.001$).

4.5 `NLPContributions-Auto`: A corpus of Auto-Identified Contribution Statements

We apply the fine-tuned SciBERT model to the abstracts of papers from ACL Anthology and classify them according to the predefined taxonomy. We call this corpus `NLPContributions-Auto`. This corpus can be used for diverse research purposes on NLP papers, including efficient semantic searches and key point analysis, among others. In the following section, we explore various NLP research trends using this corpus.

Specifically, we used S2ORC to gather the abstracts of 28,937 papers published from

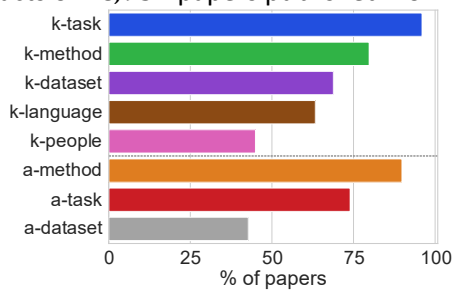


Figure 2: Distribution of contribution types across research papers.

conferences or journals falling within the "ACL Events" category between 1974 and February 2024
(details in Appendix B). We collected the metadata of these papers from the "anthology.bib".

5 Analyzing the Nature of NLP

We examine trends in NLP research contributions (§ 5.1), their influence on venues (§ 5.2), and citation counts (§ 5.3).

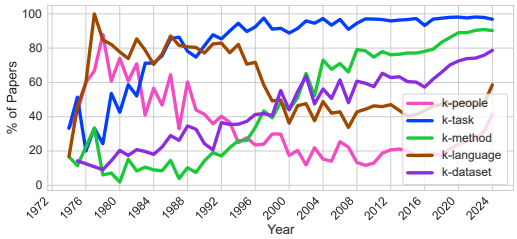
5.1 Contributions in NLP Research Papers and Evolution

Q1.What is the distribution of different contribution types in NLP research?

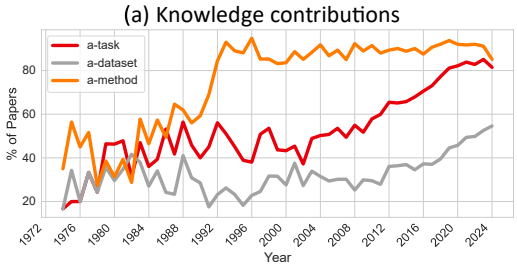
To study the breadth of contributions in NLP research, we first examine the types of contributions in `NLPContributions-Auto`. It is important to note that while NLP papers are published outside of ACL Anthology, it remains the largest single-source collection of NLP papers. Additionally, the Anthology's strict peer review process ensures high quality, making it a reliable source for our study. We identify the distinct types and sub-types present in abstracts of each paper and then aggregate these labels across the entire dataset.

Results. Figure 2 shows that contributions towards *knowledge* about people (44.9%) are less frequent compared to those concerning tasks and methods. Within the *artifact* type contributions, approximately $\sim$ 89% of the papers introduce new methods, the highest among artifact categories, followed by tasks and, finally, datasets.

While some researchers suggest that NLP research is more relevant to people or society (Schober, 1992), our observation reveals a significant focus of NLP research on knowledge about tasks and methods, particularly involving machine learning. Also, our findings resonate with those of Pramanick et al. (2023), who noted through causal entity analysis that new methods and tasks have been drivers of NLP research.



(a) Knowledge contributions



(b) Artifact contributions

Figure 3: Temporal distribution of contribution types across research papers.

Q2.How has the distribution of contribution types in NLP research papers evolved over time?

To study how the nature of NLP research has changed over time, we examine the annual distribution of the contribution labels using the same methodology described in the previous question.

Results. Figure 3a shows a marked decline in contributions towards knowledge about language or people from the early nineties through 2020 (with a drop of $\sim$40%). Figure 3b shows an increase in the number of papers proposing new methods during the same period. However, a resurgence in contributions towards knowledge about language and people is observed post-2020.

The early nineties marked a shift in NLP's focus with the advent of statistical models (Brown et al., 1993), the release of the Penn dataset (Marcus et al., 1993), and later the establishment of the EMNLP conference. While recent discussions often highlight newer methods or models (such as transformers or LLMs), our findings indicate that the shift towards contributions in methods or models began in the early nineties. That era set the stage for the development of newer methods that continue to shape NLP research. The recent resurgence in contributions toward new knowledge about people and language is attributed to an increased interest among researchers in sociolinguistics and the application of NLP in social sciences.

5.2 Contributions and Venues

Q3.How do venues influence the types of contributions made in NLP research papers published at those venues?

Each publication venue maintains distinct expectations regarding the types of work it accepts, such as the focus on particular topics, the expected degree of polish, and the nature of the experiments conducted. To study how these venues influence the contributions in papers published within these venues, we examine the distinct types and further sub-types of the contribution statements in the abstracts of the papers across different venues normalized by the number of papers published in that venue.

Results. We found that the majority of conferences (such as ACL, EMNLP, NAACL, etc.)

display similar distributions regarding the types of contributions in their published papers: roughly 68% artifacts (task: 71%, method: 89%, dataset: 42%) and 69% knowledge (task: 94%, method: 77%, people: 44%, dataset: 65%, language:61%). However, EMNLP stands out with a markedly higher number of artifact-method contributions, reflecting its focus on empirical methods. In contrast, the CL journal stands out from the rest of the *CL venues as it contributes more to the knowledge about language or people and relatively less to machine learning. We illustrate the venuewise statistics in Figure 6 in Appendix C.

Q4.How have the types of contributions in NLP research papers evolved across different venues over time?

We hypothesize that as a field matures, latent community norms emerge that influence the overall research direction, leading to a more uniform distribution of contributions across different venues over time. To investigate this hypothesis, we analyze how the specific types and sub-types of contributions documented in the abstracts of papers from each venue change over time.

Results. Among the venues we analyze, CL is the oldest journal, and ACL is the oldest conference. Interestingly, we first observe a decline in contributions concerning knowledge about people and language at the ACL, which then gradually appears in CL. This trend towards similar contribution distributions is also evident in newer conferences aligning with ACL. The temporal distribution of contribution types across these venues is illustrated in Figure 7 (Appendix C).

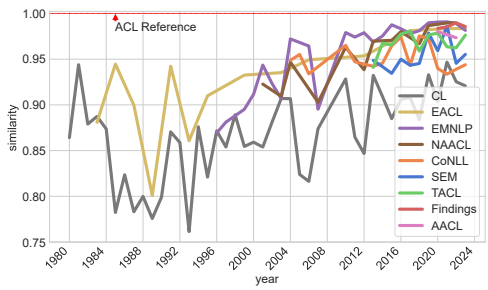


Figure 4: Comparison of venue similarity based on contribution types.

To determine if these venues increasingly reflect ACL's contribution distribution, we investigate the next question.

Q5.Are other venues increasingly resembling the ACL Conference in terms research contributions?

The ACL conference is the largest and arguably most prestigious among the ACL Events, hosting about 30.3% of the papers published in these venues. Given its prominent position, it is interesting to investigate whether other conferences have gradually begun to mirror the distribution of the types of contributions featured in ACL over time. We compare the distribution of types and sub-types of contributions in papers from these venues with that of those from the ACL conference in the same year, using the Jensen-Shannon divergence (Menéndez et al., 1997), where a value close to 1 indicates similar distributions.

Results. Figure 4 shows that over time, conferences have become increasingly similar to the ACL conference in terms of distributions of the types of contributions their papers make. For instance, the EMNLP conference, originally established to focus on empirical findings, has shown growing similarity to ACL. Similarly, newer venues like AACL and Findings closely align with ACL's contribution patterns.

This trend tends to confirm our hypothesis that, over time, a common publication norm has emerged across conferences, leading to a more institutionalized standard in NLP research. On the other hand, it is also arguably a loss that the different venues do not have unique characteristics, championing and valuing different kinds of works. *Q6.Do journal papers exhibit a greater variety of contribution types than conference papers?* Different publication venues have varying constraints on the number of pages they allow, with journals typically offering more space compared to the stricter page limits at NLP conferences. To investigate whether journal papers utilize

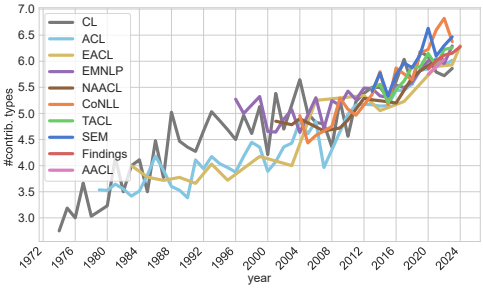


Figure 5: Average no. of contribution types per paper across different venues.

the additional space to include a wider range of contribution types, we analyze the average number of unique contributions per paper across various venues on an annual basis.

Results. Figure 5 shows that conferences and journals have similar average number of unique contribution types in their abstracts, but this number has been increasing over time. However, we observe that the average length of abstracts in venues has not changed significantly over time (Figure 8, Appendix C). The results indicate the growing sophistication and expansion of NLP applications.

5.3 Contributions and Citation Impact

Q7.Do different types of research contributions have differing amounts of citation influence? All contribution types are important for a thriving and vibrant ecosystem of NLP research. Thus, marked disparities in citation counts could potentially disincentivize work on certain types of contributions. Therefore, through this question, we track the citational impact of different contribution types. We calculate the average and median citation counts for each type of contribution from the papers that have at least one contribution statement pertaining to that type. To ensure a meaningful assessment of citation trajectories, we focus on papers with at least five years of publication history (Anderson et al., 2012). For this purpose, we selected 352 papers from the ACL'18 to examine the citation impact of papers published simultaneously for this experiment. Results. Table 5 shows that the papers introducing new datasets receive notably high citations. Additionally, papers proposing new methods attract more citations than those proposing new tasks. Papers that add knowledge of methods or datasets (primarily through analysis) receive more citations than those that add knowledge about people or language, indicating the community's

Contribution		#papers	#citations	
Typ.	Sub-typ.		mean	median
Knowledge	k-dataset	219	121.1	56.0
	k-language	193	107.1	53.0
	k-method	280	127.8	56.0
	k-people	119	109.5	54.0
	k-task	328	115.7	55.0
Artifact	a-dataset	154	137.7	64.0
	a-method	310	122.2	58.0
	a-task	270	116.0	56.0

Table 5: Mean and median citation counts of papers for different contributions in ACL'18.

stronger interest in technical advancements over sociolinguistic studies. It is notable that even though there are more papers adding to the knowledge about language than with a focus on adding knowledge about people, the languagecontribution papers tend to receive fewer citations. Further analysis of 277 papers from ACL'17 (Table 9, Appendix C) shows similar trends.

It is important to recognize that citations are influenced by various factors beyond just contribution types. Our objective is neither to identify all possible influences on citation counts nor to pinpoint the most influential factors. However, the high citations for papers that create new datasets perhaps reflect the importance of datasets in much of NLP research, particularly for training and evaluating models - a common practice in modern NLP.

6 Concluding Discussion

In this paper, we proposed that automatically extracting, categorizing, and quantitatively analyzing the contribution statements in the research papers of a field provides key insights into the nature of the field. We first established a taxonomy of contributions and developed a systematic framework to automatically extract, classify, and analyze contribution statements from NLP research papers. Our analysis revealed that although NLP is intrinsically linked to

linguistics and society, research focusing on language and people has waned since the early nineties, while contributions to developing methods, mainly using machine learning tools, have risen. It is important to note, however, that an increase in methodological contributions does not necessarily imply a diminished focus on language or people; these areas can expand simultaneously. Recently, post-2020, there has been a revived interest in sociolinguistics and the application of NLP within social sciences, suggesting a broadening of research focus. Additionally, while the NLP community initially created various publication venues to address different research types—such as EMNLP for empirical investigations—the distinctions between these venues have diminished over time, indicating a growing uniformity in the community.

Although the growth of NLP is beneficial, we emphasize the importance of maintaining diversity in research contributions to ensure the field remains relevant to a broader community. As members of this community, we hold the strength to guide the future direction of these trends. However, we are not advocating for a specific stance on research practices but encourage an inclusive approach that embraces a variety of contribution types within NLP research. To foster future research in the area of contribution analysis and stimulate informed discussions within our community, we release our artifacts under *cbna*.

Limitations

This study primarily examines NLP research papers from the ACL Anthology, specifically focusing on papers from conferences and journals under ACL Events, such as ACL, EMNLP, NAACL, EACL, and journals like TACL and CL. However, it is crucial to recognize that significant NLP research also appears outside the ACL Anthology, including in AI venues, regional conferences, and preprint servers. While papers published in the ACL Anthology are typically of high quality, research from other venues often contributes valuable insights to the field. We leave the effort to curate and include research papers from these alternative venues for future work.

Our study primarily analyzes the abstracts of research papers, which are typically concise, logically coherent paragraphs that hold a unique position within the paper. While abstracts are likely to contain the key contributions as highlighted by the authors, making them a focal point for initial analysis, it is important to acknowledge that unique contributions may also be found within the main body of the paper. However, annotating the full text of research papers requires significant time, effort, and substantial domain knowledge to accurately understand and contextualize the content. In future iterations of this work, we plan to extend our annotations to include the main body of the papers, providing a more inclusive dataset.

Finally, for our analysis, we first train a classifier on the high-quality, human-annotated dataset that we create and then deploy this trained model on the larger ACL Anthology dataset. We conduct our analysis based on the labels generated by this model. It is important to acknowledge that no model achieves perfect accuracy, which can impact the quality of such analysis. However, as demonstrated by Teodorescu and Mohammad (2023), when broader cumulative trends are derived from large datasets using such models, the results tend to be highly accurate and show a strong correlation with trends identified through goldlabel analysis. This supports the reliability and accuracy of our analysis despite the inherent limitations of trained machine-learning models.

Ethics Statement

In this work, we utilize publicly accessible data from the ACL Anthology and do not involve any personal data. It is important to acknowledge that although our approach is datadriven, individual views on research are naturally subjective. Therefore, decisions in science should not only be based on data but also take into account ethical, social, and other qualitative considerations.

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A `NLPContributions` Annotation Guidelines

We propose a linguistic annotation scheme to study and analyze the types of contributions articulated in NLP research papers following the taxonomy we developed in § 3.1. The goal of this annotation scheme is to annotate contribution statements from the abstract section of NLP research papers into various types and further into sub-types as mentioned in Table 3, according to the specific aspect of field advancement they represent. In this section, we elaborate on both broad categories and more detailed sub-types of contributions within each category.

A.1 Contribution Types

A.1.1 Type: Artifact

This type of contribution includes the creation of new resources such as datasets, models, or algorithms. These artifacts play a crucial role in NLP research, which frequently uses tools from Machine Learning. We further sub-categorize contributions into three distinct categories based on the type of artifact they introduce to the field.

- *a-tasks:* Researchers often identify or formulate new or previously unknown problems (such as linguistic problems like NER Tagging) and formally describe them in research papers as tasks.

- *a-Methods:* Researchers often create new NLP methods such as algorithms or models (for example, BERT (pre-trained language model) or LLaMA (large language model)) as artifacts primarily to solve tasks and describe them in research papers.

- *a-dataset:* Researchers create new scientific corpus or language resources as artifacts to build models or analyze languages, such as SQuAD or Penn Treebank.

A.1.2 Type: Knowledge

This type of contribution encompasses the addition of new insights or understandings to the field. These contributions often relate to linguistic studies due to the significant overlap between NLP and linguistics, or they may explore societal and human aspects because of NLP's focus on human language. Consequently, we further categorize it into five subtypes based on the specific area of knowledge it expands.

- *k-task:* Contribute new insights into specific NLP task(s).
- *k-dataset:* Contribute new insights or analysis of an NLP dataset.

- *k-method:* Enhances the understanding of algorithms, methods or models within NLP.

- *k-language:* Adds new knowledge about natural language.

- *k-people:* Explores and adds knowledge about aspects of human behavior and social implications as revealed through natural language.

A.2 Annotation Instructions

First, we present the following two definitions to the annotators along with the contribution types as described in § A.1.

Definition A.1 (Contribution). A contribution is a scientific achievement attributed to the authors of a research paper, such as introducing a new model or dataset.

Definition A.2 (Contribution Statement). A statement in a research paper that describes new scientific achievements attributed to its authors is called a contribution statement.

Next, we present the annotators with the title and abstract of a research paper and instruct them to annotate each statement of the abstract according to the following questions.

Q1. *Does the sentence qualify as a contribution statement according to the definitions presented earlier?*

In the second question, multiple options could be selected.

Q2. *If you answered yes to the previous question, which of the following options most accurately describes the type and sub-type of the contribution statement? Please select all that apply.*

- *Artifact-Task* (Introduces, proposes, or formulates a new or novel NLP task.)
- *Artifact-Method* (Introduces or creates a new or novel NLP method such as an algorithm or a new NLP model.)
- *Artifact-Dataset* (Creates a new corpus or language resource.)

Typ.	Sub-typ.	#Contrib.
Knowledge	k-dataset	25
	k-language	23
	k-method	53
	k-people	41
	k-task	122
Artifact	a-dataset	20
	a-method	94
	a-task	22

Table 6: The number of Labels in the Annotated Contribution Statements by the two annotators.

- *Knowledge-Task* (Describes new knowledge about NLP task.)
- *Knowledge-Dataset* (Describes new knowledge about datasets, such as their new properties or characteristics.)
- *Knowledge-Method* (Describes or presents new knowledge or analysis about NLP models or methods, which are primarily drawn from Machine Learning.)
- *Knowledge-Language* (Presents new knowledge about language, such as a new property or characteristic of language.)
- *Knowledge-People* (Presents new knowledge about people, humankind, society, or human civilization.)
- Others (Any other type that does not fall under the categories mentioned above.)

Discussion: We observe that only a small number of NLP papers propose new *metrics for evaluation measures*. Since these metrics function as algorithms or methods, we categorize them under the class "artifact-method".

Also note that, in the initial pilot annotation study, we included an "Others" label (as mentioned above) to capture any types of contributions not already accounted for in our taxonomy. This allowed us to potentially expand our taxonomy based on the pilot results. Following the pilot study, however, we found that our existing taxonomy adequately covered all types of contributions identified by the annotators.

A.3 Annotation Statistics

We provide statistics for the abstracts of 100 papers annotated by two annotators post-adjudication. Of the 584 statements annotated, 359 were identified as contributions. Table 6 details the number of these statements across each type and subtype of contributions.

B Corpus Details

To address the question *"What constitutes NLP Research?"*, first We used S2ORC to gather the abstracts of $29{,}010$ papers published from conferences or journals falling within the "ACL Events" category. Notably, in 1997, the ACL and EACL conferences were held jointly, resulting in 73 papers being listed under both events for that year. We treated these dual-listed papers as single entries, reducing our dataset to a total of $28{,}937$ unique research papers. Additionally, we collected the metadata associated with these papers from the "anthology.bib".

Finally, we applied the SciBERT model, finetuned on `NLPContributions`, to these abstracts to identify the contribution statements and classify them according to the predefined taxonomy.

C Additional Results

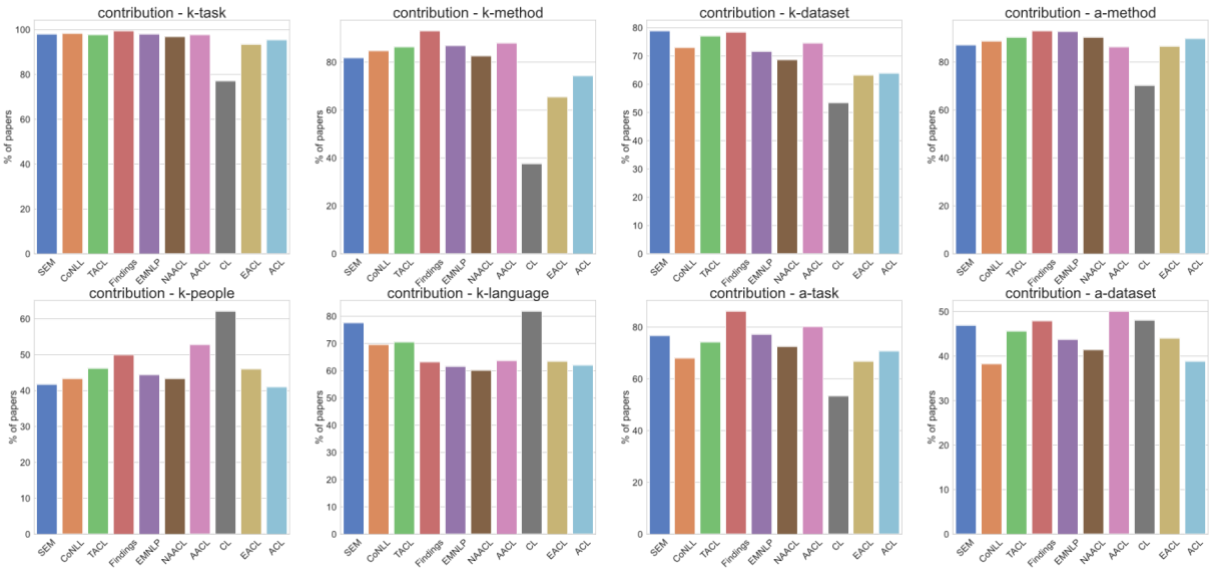

Figure 6: Distribution of contribution types across research papers by conference (Abbr.: knowledge (k), artifact (a)).

Type	Sub-type	Prompt
Knowledge	k-task	Central to NLP research are tasks such as Machine Translation, Named Entity Recognition, Language Modeling, etc. Your task is to assess whether the provided sentence from an NLP research paper describes new knowledge about any of such existing NLP tasks, including new knowledge about their properties or characteristics. However, the sentence should not propose a new NLP task. Respond with "yes" if the sentence presents new knowledge about one or more of these NLP tasks; otherwise, respond with "no".
	k-method	NLP Models such as RNNs, LSTMs or LLMs are indispensable for NLP Research. Your task is to determine if the provided sentence from an NLP research paper describes new knowledge or analysis about such existing NLP models or methods like RNNs, LSTMs, or LLMs. However, the sentence should not propose new models or methods. Respond with "yes" if the sentence presents new knowledge about NLP models; otherwise, respond with "no."
	k-people	In NLP research, every paper plays a role in advancing the field. Your task is to assess whether the given sentence from an NLP research paper presents new knowledge about people, humankind, society or human civilization. Respond with "yes" if the sentence describes novel knowledge about people, humankind, society or human civilization; otherwise, respond with "no." Use only a yes or no format for your answers.
	k-dataset	Datasets constitute a crucial aspect of NLP and machine learning research. Examining datasets can yield valuable insights into their properties and features. Your task is to assess whether the given sentence from an NLP research paper describes new knowledge about a dataset, such as its new properties or characteristics or describes new knowledge concerning properties or characteristics of datasets in general. Respond with "yes" if the sentence presents novel knowledge about the datasets; otherwise, respond with "no." Use only a yes or no format for your answers.
	k-language	In NLP research, every paper plays a role in advancing the field. Your task is to assess whether the given sentence from an NLP research paper presents new knowledge about language, such as a new property or characteristic of language. Respond with "yes" if the sentence describes novel knowledge about language; otherwise, respond with "no." Use only a yes or no format for your answers.
Artifact	a-task	Central to NLP research are tasks such as machine translation, named entity recognition, sentiment classification, and more. Your task is to assess if the given sentence from an NLP research paper introduces, or proposes a new or novel NLP task. This new task could either build upon existing NLP tasks or could be entirely novel. Respond with "yes" if the sentence introduces, or proposes a new or novel NLP task; otherwise, respond with "no."
	a-method	Algorithms and NLP models such as RNNs, LSTMs or LLMs are indispensable for NLP Research. Your task is to assess if the provided sentence from an NLP research paper introducing, or proposing a new or novel such NLP model, algorithm, or technique. This new model could have been built on top of existing models or methods or could be a completely new model. Respond with "yes" if the sentence introduces or proposes a new or novel NLP model; otherwise, respond with "no."
	a-dataset	Datasets constitute a crucial aspect of NLP research. Your task is to assess whether the given sentence from an NLP research paper introduces or discusses a new or novel NLP dataset. Respond with "yes" if it does; otherwise, respond with "no."

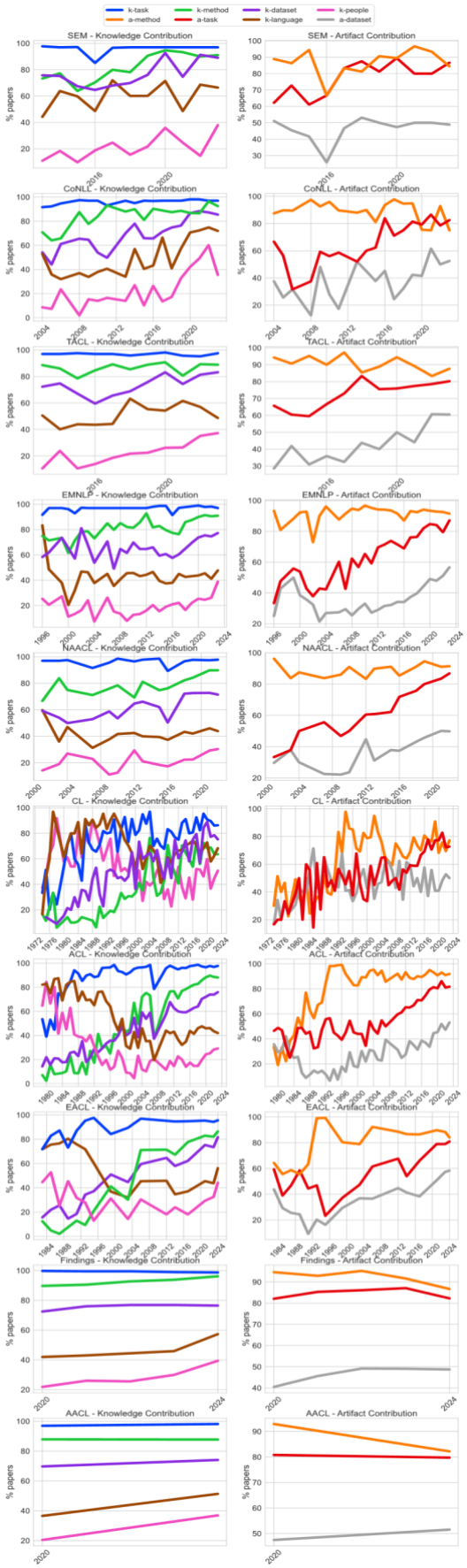

Figure 7: citation counts by contributions for ACL 2017 Papers.

Table 7: Prompts to identify different types of contributions from NLP Research papers using LLMs.

Table 8: LLM performance comparison using Macro F1 Score with different number of training examples.

Contribution		#papers	avg. citation (↑)
Typ.	Sub-typ.		
Knowledge	k-dataset	180	120.0
	k-language	160	110.3
	k-method	230	135.3
	k-people	101	104.7
	k-task	254	121.5
Artifact	a-dataset	96	140.7
	a-method	250	131.5
	a-task	202	120.8

Table 9: Avg.

venue		avg. #sent.
conference	ACL	5.69
	EMNLP	6.23
	NAACL	5.74
	EACL	5.67
	AACL	6.43
	Findings	6.94
	SEM	5.48
	CoNLL	5.73
journal	TACL	6.10
	CL	9.01

Table 10: Average number of sentences in abstracts for papers published across various conferences.

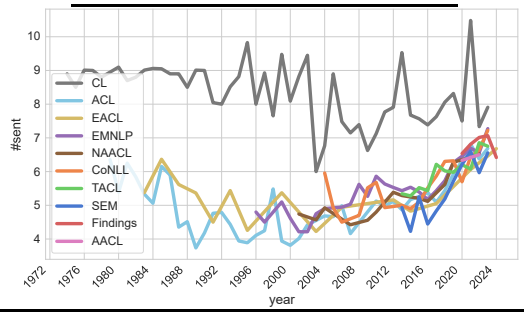

Figure 8: Average abstract length of per paper across

Model	0-shot	1-shot	3-shot	5-shot
GPT-3.5-Turbo	0.52	0.55	0.64	0.73
GPT-4-Turbo	0.61	0.66	0.72	0.80
LLaMA-3	0.50	0.50	0.51	0.53

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nt in the distribution of different venues.

contribution types across research papers by conference (Abbr.: knowledge (k), artifact (a)).